

In this Chapter you will learn about:

- Transfer of Heat
- Modes of transfer of heat (Conduction, Convection and Radiation)
- Everyday applications (Conduction of heat, convection currents and radiation).
- Good and bad conductor of heat.
- Ocean currents.
- Good and bad radiators of heat.
- Good and bad absorbers of heat and vacuum flask.

All the students will be able to:

- ✓ Explain the flow of heat from hot body to cold body.
- ✓ Explain conduction, convection and radiation through experimentation.
- ✓ Recognize three modes of transfer of heat from environment.
- ✓ Suggest how birds can glide in the air for hours.
- ✓ Identify examples of appliances that make use of different modes of transfer of heat.
- ✓ List heat conducting materials in their surroundings.
- ✓ Describe the working and principle of vacuum flask.
- ✓ Explain how a vacuum flask reduces the transfer of heat.

Have you ever thought why do we feel hot during a sunny day? Do you feel hot while holding a metal spoon during cooking? Why do we see bubbles in water, while it is boiling? The answer is that heat transfers from one place to another, having different temperatures.

Modes of heat transfer in above examples are radiation, conduction and convection. Can you name the materials used to avoid heat transfer in above examples?



Figure 8.1 Transfer of heat

TRANSFER OF HEAT

- ✓ Explain the flow of heat from hot body to cold body.

As you have studied in previous classes that matter is made up of atoms and molecules, which are always in motion - either bumping into each other or vibrating back and forth. The motion of atoms and molecules results in the formation of energy called heat energy.

Often we think that heat and temperature are the same thing but you have studied in class six that, this is not the case. Heat and temperature are related to each other, but are entirely different quantities. Heat is the total energy of molecular motion in a substance while temperature is the degree of hotness or coldness measured by a definite scale. Heat energy can be transferred from one object to another, and the transfer is due to the difference in temperature between the two objects.

Activity 8.1: Investigating the transfer of heat.

What I need:

- Hot water (boiled, almost of 100°C temperature)
- Cold water at 10 to 15°C
- Three containers (beakers)
- Thermometer
- Measuring cylinder

What to do:

- First note temperatures of hot and cold water separately, with the help of thermometer.
- Take 100ml of cold water in third container and add 50ml of hot water into it and note the temperature.
- In the same container further add 50ml (total 100ml) of hot water and note the temperature.
- Again in the same container, further add 50ml (total 150ml) of hot water and note the temperature.

What I observed:

State of Water	Temperature in $^{\circ}\text{C}$
Cold water	
Cold water 100ml + 50ml of hot water	
Cold water 100ml + 100ml of hot water	
Cold water 100ml + 150ml of hot water	

Activity Questions:

1. What was the temperature of hot water?
2. What was the temperature of cold water?
3. What was the temperature of cold water after adding 50ml of hot water?
4. Did you notice change in temperature of cold water while you kept on adding hot water into it?

What I concluded:

When two bodies of different temperatures come into contact by any means, they either lose or gain heat from each other, till both come to the same temperatures. Losing or gaining heat due to difference of temperatures in two bodies or systems is called transfer of heat. It is not necessary for the bodies to come in physical contact for transfer of heat. Heat can be transferred by other means as well, such as hot currents (in fluids) and heat waves in air and space. In the later situations, generally either the bodies or systems do not reach to the same temperature due to involvement of some other factors, for example.

MODES OF HEAT TRANSFER

- ✓ Explain conduction, convection and radiation through experimentation.
- ✓ Recognize three modes of transfer of heat from environment.

Heat transfers from an object of high temperature to that of a lower temperature. There are three ways in which heat is transferred from one place to another, called modes of heat transformation.

Modes of heat transformation:

- Conduction
- Convection
- Radiation

Activity 8.2: Conduction of Heat

What I need:

- Iron rod
- Candle
- Burner/spirit lamp
- Match box

What to do:

1. Mark four points on iron rod at equal distances.
2. Lit the candle and put some drops of melting candle wax on iron rod at marked points.
3. Let it cool in order to form wax balls on iron rod.
4. Now lit the burner/spirit lamp and start to heat up the iron rod from the end having wax balls.
5. It is not safe to hold the rod while heating it. Tie the rod with a string or use a clamp.
6. As you start to heat it up, note down the time.
7. As the first wax ball gets melt, note down the time and so on.
8. Calculate the interval time to melt each two balls and time to melt all the balls.

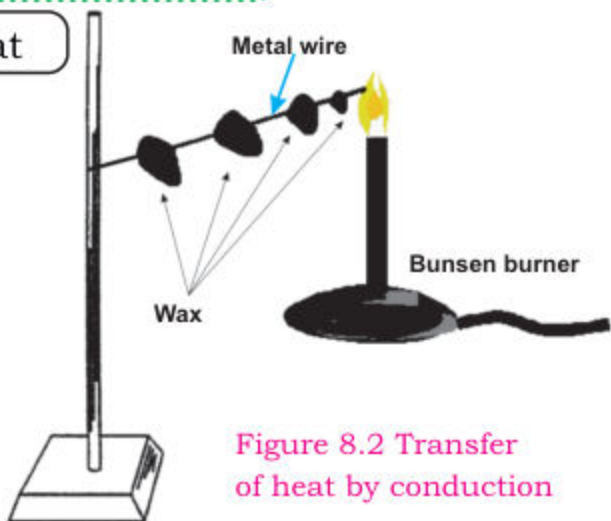


Figure 8.2 Transfer of heat by conduction

What I observed:

State of Wax Ball	Time	Difference of time to melt two balls in (sec)
Initial time		
First wax ball melts		
Second wax ball melts		
Third wax ball melts		
Fourth wax ball melts		
Total time taken to melt all the balls		

Activity Questions:

1. How much time was taken to melt all the wax balls?
2. Why did the wax balls melt?
3. What did you understand by melting of wax balls?
4. Can you name this type of transfer of heat?

What I concluded:

Conduction is one of the modes of transfer of heat. Matter is made up of atoms or molecules, very close to each other and tightly packed in solids. In solids they move around fixed positions in the form of vibration. When molecules get heat from any source, they become hot.

Molecules transfer heat to the neighbouring molecules having a less temperature, through their vibrating movements. Heat transfers from one molecule to another and so on, resulting in heat transfer from one end to another. Conduction occurs only conductor in solids.

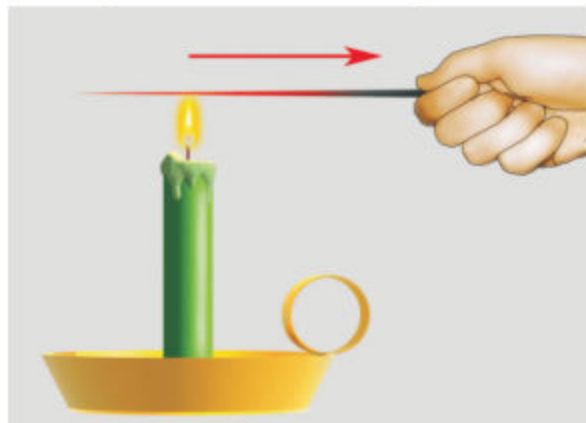


Figure 8.3 Heat transfers by conduction

Everyday application of conduction of heat:

1. When we cook something using a pot or a frying pan, we are using the pan to transfer heat from the stove to the food.
2. When we get sick and use a mercury thermometer to measure temperature, we are using the metal tip of the thermometer to transfer our body's heat to the mercury.
3. To hold a frying pan we use a wooden or plastic handle to save our hands from heat being transferred by conduction.

Teacher Note: Help students in making groups. Also instruct them to take precautions while dealing with fire and observe carefully while they take readings. Wax ball should be at the distance from first end.

GOOD AND BAD CONDUCTORS OF HEAT

We know that conduction only occurs in solids and transfers heat from high temperature to lower temperature. However, there are some solids which do not conduct heat. The solids that conduct heat are called as good conductors of heat and those which do not conduct heat are called as bad conductors of heat or insulators. Mostly, metals are good conductors of heat. Copper is considered as the best conductor. Some materials like wood, plastic and paper etc are bad conductors of heat.

Think about

- What kind of material is used for pan to cook food?
- How woollen clothes keep us warm in winter?

Activity 8.3: Demonstrating the convection currents in liquids.

What I need:

- Coloured beads/plastic pieces/ Paper pieces/colour pigment
- Beaker
- Water
- Wire gauze
- Tripod stand
- Burner/spirit lamp
- Match box

What to do:

1. Get apparatus set in order. Put tripod stand over the spirit lamp, wire gauze on tripod stand and put beaker on wire gauze.
2. Pour water into beaker.
3. Now lit the spirit lamp and let the water boil.
4. As water is about to boil place coloured beads/plastic pieces/ paper pieces into water and observe its movement in water.



Figure 8.4 Transfer of heat by convection

DO YOU KNOW?

Conductors

Iron
Copper
Silver
Mercury
Aluminium

Insulators

Wood
Plastic
Wool
Cork

Teacher Note: Help the students to observe in groups or make it visible for all the students. Help them to understand the movement of coloured beads, plastic pieces, paper pieces. Instruct them to take precautions while they come closer to the apparatus.

Activity Questions:

1. What did you observe?
2. Why are the coloured beads /plastic pieces/paper pieces moving in the beaker of water?
3. How are the coloured beads /plastic pieces/paper pieces moving?

What I conclude:

Convection is another mode of heat transfer. Unlike conduction, it occurs in liquids and gases. As their molecules move freely unlike solids. Convection occurs due to the movement of fluids' molecules. When the molecules become hot at the bottom, they become lighter and rise up. Comparatively colder molecules move downwards to fill up the gaps in the bottom. In this way convection currents are formed in liquids and the same process happens in gases.

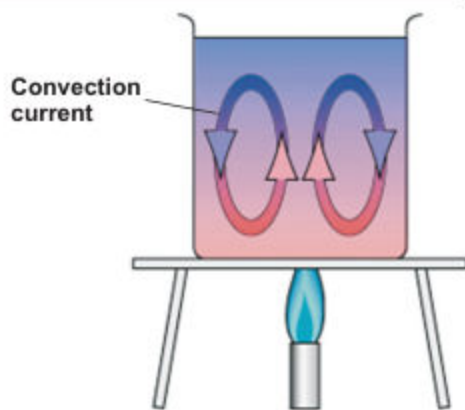


Figure 8.5 Heated water rises from the bottom to the top of the pot. Cold water replaces the rising water

EVERYDAY APPLICATION OF CONVECTION OF HEAT

1. Land and Sea Breeze:

Water and land have different heating abilities. Water takes a bit more time to warm up and is able to retain the heat for longer time than land does. Due to the convection in atmosphere, two types of breezes occur along coastal areas or areas adjacent with large water bodies.

In the day, when the sun is up, the land heats up very quickly and the air above it warms up more than that above the water. The warm air over the land becomes less dense and rise up resulting to create low pressure.

The air pressure over the water is higher with cold dense air, which moves to occupy the space created over the land. The cooler air that comes along is called sea breeze.

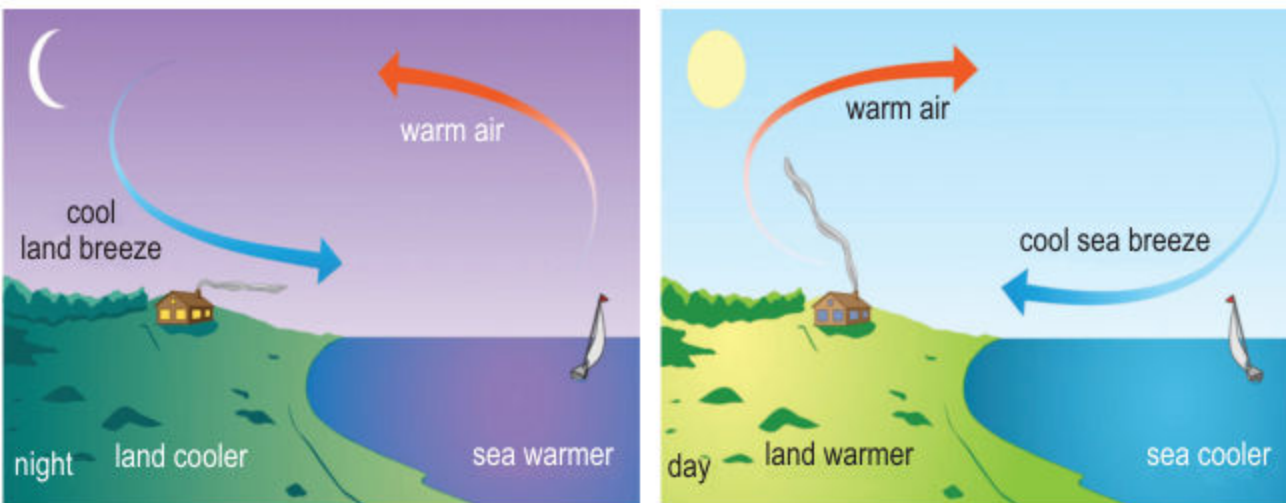


Figure 8.6 Land breeze and sea breeze

At the night, the reverse happens. The land quickly loses its' heat while water does not. This means the air over the sea water is warm, less dense and begins to rise. Low pressure is created over the sea. Cold and dense air over the land begins to move to the water surface to replace the warm rising air. The cool breeze from the land is called land breeze.

2. Ocean Currents and Winds:

Ocean currents can occur on local and global scales and are typically wind-driven, resulting in both horizontal and vertical movement of water. An ocean current is a continuous and directed movement of sea water. Ocean currents flow for great distances. Wind driven ocean currents are effected by many external factors including Earth's rotation, temperature, salinity and gravitational pull.

Global wind patterns and ocean currents determine the climate and weather of the world. Convection, which is the transfer of heat between liquids and gases, occurs between the ocean water and the air, giving the air a varying degrees of temperature and it forms climate.

Activity 8.4: Radiation of Heat

What I need:

- Tin can
- Laboratory thermometers
- Hot water

What to do:

1. Pour hot water in tin can.
2. Insert one thermometer in tin can.
3. Note down the temperature of water in the tin can after the given intervals.



Figure 8.7 Transfer of heat by radiation

What I observed:

Intervals	Temperature in $^{\circ}\text{C}$
Initial	100 $^{\circ}\text{C}$
After 5 minutes	
After 10 minutes	
After 15 minutes	
After 20 minutes	

Activity Questions:

1. Did you notice any change in temperature?
2. To what extent, temperature is decreased after 20 minutes?
3. What was heat transferred to?
4. Can you name this process of transfer of heat?

What I conclude:

Radiation is the transfer of heat through waves. To *radiate* means to send out or spread from the source of heat. The transfer of heat by radiation involves the carrying of heat energy from an origin to the space surrounding it. The heat is carried by waves and does not involve the movement or the interaction of matter. The hotter the object, the more it radiates. The heat received on earth from the sun is the result of radiation.

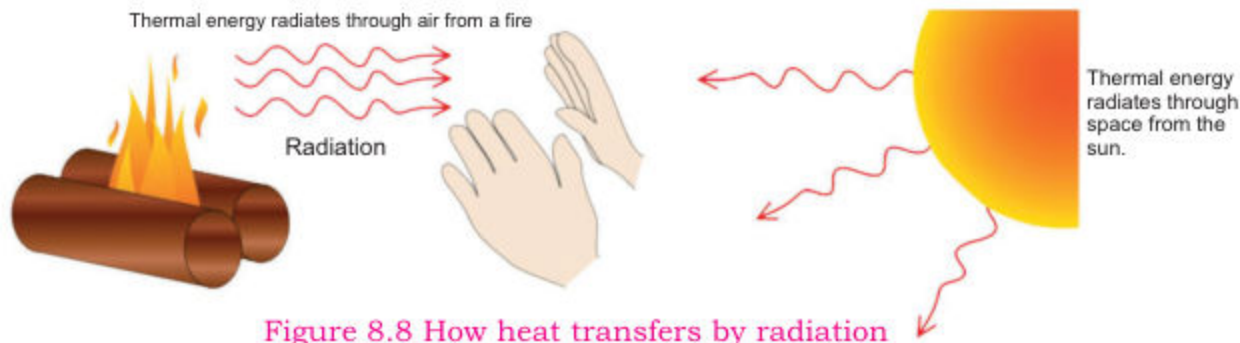


Figure 8.8 How heat transfers by radiation

GOOD AND BAD ABSORBERS AND RADIATORS OF HEAT

The rate at which an object radiates or absorbs heat by radiation depends on its temperature and nature of its surface. The hotter the object is, the more heat it radiates. Some surfaces are better radiators of heat than others. Most metals are good conductors of heat; silver and copper are exceptionally good. While the substances such as cork, wood, cotton and wool are bad conductors of heat. Dull black surfaces are good absorbers and good radiators of heat. On the other hand, shiny surfaces are poor absorbers and poor radiators of heat; they reflect the radiations away.

Everyday application of radiation of heat:

1. We receive heat from sun through radiation.
2. We feel warm in winter while sit near the fire place.
3. People of hot countries use light colours for their houses.
4. Water remains hot longer in shiny kettle as it is bad radiator of heat.
5. Shiny or white petrol tanks reflect heat from the sun and prevent petrol from heating up.

Think about

Fatima is a nature lover, she feels herself as a part of nature and enjoying all the seasons. She is always conscious about wearing dresses as per the seasons. She selects beautiful colours for her dresses that enhance her mood to enjoy nature. She always selects light colours for summers and dark colours for winter.

Can you comment that why she avoids dark colours in summers?

✓ Suggest how birds can glide in the air for hours.

The process of convection also occurs in atmosphere. In the day, when the sun is up, the land heats up very quickly and the air above it warms up a lot. Mostly in the afternoon, the warmer air near the



Figure 8.9 Bird is gliding in air due to thermal (Hot Air)

earth's surface expands, becoming less dense than the surrounding air. The lighter hot air rises (thermals) and get cool at higher altitude. It stops rising when it has cooled to the same temperature as the surrounding air. Birds use thermals (warm light air), their wings are useful for making use of rising warm air (due to convection). The air travels faster above the bird's wings than it does below; resulting it's flying in air.

- ✓ Identify examples of appliances that make use of different modes of transfer of heat.
- ✓ List heat conducting materials in their surroundings.

1. Refrigerator:

Refrigerator is a commonly used appliance to keep food cool to keep it for longer time. In the refrigerator, the compressor at the back, draws air through the evaporator fins to cool the food, by convection. That heat from the food is transferred into the refrigerant through the tube walls from the fins by conduction. That heat is compressed through condenser tubes and fins; transferred into the room which is carried away. There is a small amount of heat reduced by radiation. As the refrigerator needs to radiate its heat to the surroundings.



Figure 8.10 Refrigerator

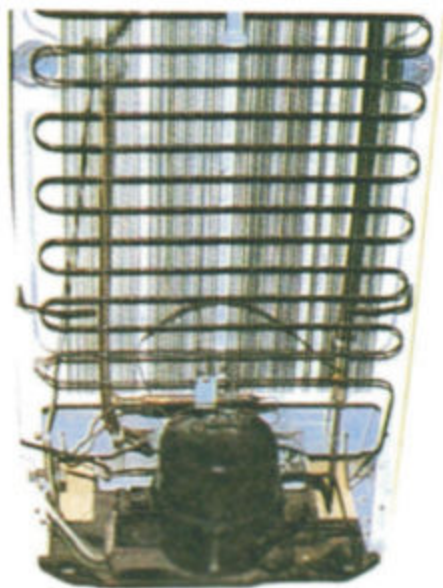


Figure 8.11 Back fins of refrigerator

2. Air Conditioner:

Air conditioners and refrigerators work the same way. Instead of cooling just the small, insulated space inside of a refrigerator, an air conditioner cools a room or a whole house.

The cooling unit of air conditioner is placed at a height inside the room. The warm air in the room rises up to the cooling unit it gets cooled thus the room gets cooled.



Figure 8.12 Air Conditioner at the top

THE VACUUM FLASK

- ✓ Describe the working and principle of vacuum flask.
- ✓ Explain how a vacuum flask reduces the transfer of heat.

The vacuum flask is a special kind of bottle in which hot liquids like tea and milk remain hot and the cold items like ice and cold water remain cold for a long time. It consists of a double walled glass bottle. These walls are silvered from inner side. The space between the walls is evacuated with the help of a vacuum pump and sealed in order to create vacuum.

The vacuum flask is to prevent, the flow of heat to and from the bottle. We know that heat can flow from one place to another by three modes- conduction, convention and radiation.

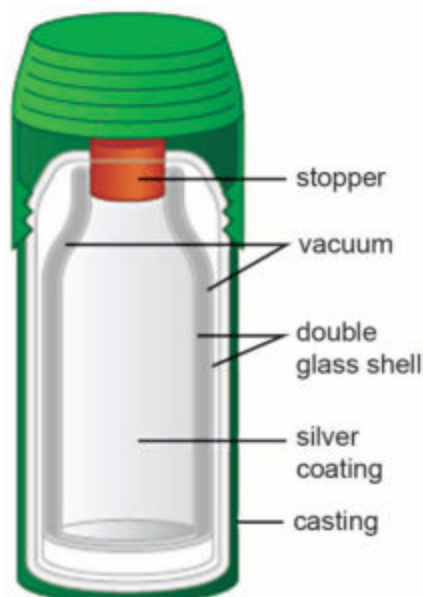


Figure 8.13 Internal structure of vacuum flask

A vacuum flask prevents the flow of heat by any of these modes. Since there is a vacuum between the walls of the bottle, which controls flow of heat by convection and conduction. Further the silvered walls stop the loss of heat by radiation. The combination of a vacuum and the silvering lining of glass greatly reduce heat transfer by conduction, convection and radiation.

As a result, hot things kept in a thermo flask or vacuum flask do not become cold and cold ones do not become hot for a long time.

Summary

- Heat is a form of energy.
- Heat can be transferred from one object to another; from high temperature to the lower temperature.
- There are three modes of heat transfer that is conduction, convection and radiation.
- Conduction does occur only in solids; metals are the good conductors of heat.
- The objects which do not conduct heat, are called bad conductors or insulators such as wood, plastic, glass etc.
- The process of convection occurs in liquids and gases.
- Heat transferring by radiation does not need any medium.
- Convection occurs in atmosphere due to which sea and land breezes are blowing. It also helps the birds to glide in air for hours easily.
- We receive sun heat through the process of radiation.
- Different colours radiate and absorb heat at different rate.
- Devices are being invented on the phenomenon of heat transfer such as refrigerator, air-conditioned etc.
- Vacuum flask is the invention most commonly used in our daily life to reduce heat transfer by any mode.

Review Exercises

1. Match the statement of column 'A' with column 'B'.

Column A	Column B
i) Degree of hotness and coldness of any object	a) White colour
ii) Best conductor of heat	b) Copper
iii) Wind driven currents	c) Radiation
iv) Transfer of heat in vacuum	d) Temperature
v) Poor radiator of heat	e) Ocean currents

2. Fill in the blanks.

- i) In the evenings cool breeze from the land to sea is called a _____.
- ii) Refrigerator works on the phenomenon of _____ of heat transfer.
- iii) An ocean current is a continuous, directed movement of _____.
- iv) We do receive sun heat by the process of _____.
- v) Insulators do not _____ heat.

3. Explain why:

- i) We use vacuum flask to keep hot tea remains hot.
- ii) Conduction occurs only in metals.
- iii) Why Sun's heat does not reach the earth through conduction or convection?

4. State five application of radiation in our daily life.

PROJECT

Do all Colours Absorb Heat Better?

Leave ice cubes placed in boxes made of coloured paper (one box per colour; white, yellow, red and black) in the sun, and predict in which coloured box ice cubes melt first. Record the order and time required for the ice cubes to melt.

ACTIVITY SHEET

Name _____

Date _____

Do different colours absorb heat better than others?

Colours	Time to Melt (min)
White	
Yellow	
Red	
Black	

In which box ice cube melted first? _____

In which box ice cube melted second? _____

In which box ice cube melted third? _____

In which box ice cube melted fourth? _____

Do different colours absorb heat better than others? _____

Investigating Questions:

- Why do ice cubes melt?
- How does the sun affect ice?
- On which colour did the first ice cube completely melt? Why?
- Which colour absorbs heat the quickest in the sun?
- Which colour would be the best to help keep ice cubes from melting too quickly in the sun?

Teacher Note:

- Make enough ice cubes so that each group can have four. Try to make them the same size for experiment consistency.
- To save time, pre-cut and assemble (using tape) the coloured paper into five-sided boxes each big enough to fit an ice cube. Otherwise, have students cut, fold and tape together their own boxes.
- Gather the rest of the materials.
- Make copies of the activity sheet per group.